Structural damage identification through PCA and subspace comparison

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***Abstract*— A numerical study is presented to assess the sensitivity of a PCA-based subspace comparison method for structural damage detection. Results demonstrate that principal angles between subspaces reliably quantify damage severity, confirming the robustness of the approach under controlled conditions.**

Keywords*— *Structural Health Monitoring, Structural diagnostics.

I. INTRODUCTION

Structural Health Monitoring (SHM) represents one of the most active and challenging research areas in civil engineering, since it directly concerns the preservation, safety, and functionality of infrastructures that are inevitably affected by material degradation and aging. In particular, motorway viaducts and prestressed concrete bridge decks are highly exposed to environmental actions, traffic loading, and long-term deterioration processes, which can progressively compromise their serviceability and structural integrity. The need for reliable diagnostic tools, capable of detecting anomalies before they evolve into severe damage, is therefore widely recognized in both the scientific community and professional practice.

Within this broad field, a variety of statistical and data-driven approaches have been explored to process the large amount of information provided by experimental monitoring systems. Among them, methodologies based on Principal Component Analysis (PCA) combined with subspace comparison through principal angles have proven to be especially effective. The strength of this approach lies in its ability to condense the high-dimensional information contained in ambient vibration data into a reduced representation that captures the essential features of the structural response. Such a reduction is crucial because vibration measurements collected under operational conditions are inherently affected by environmental variability and noise, which may mask or distort the signatures of damage. By exploiting the properties of PCA, it becomes possible to filter out much of this variability and to focus on the structural characteristics that are most informative for diagnostics (Agrò 2022; 2025).

In previous investigations, this methodology was applied to real motorway viaducts, where it was tested under true operational scenarios. The results demonstrated its capability to reveal anomalies in the dynamic response of the structures, even when only ambient vibrations were available as input data (Agrò 2022). To further strengthen the reliability of the diagnostic indicators, the approach was subsequently enhanced through the integration of bootstrap resampling procedures for time series. This extension made it possible to assess the statistical significance of the results, thereby providing a quantitative measure of confidence and reducing the risk of misinterpretation due to random fluctuations. The combination of PCA and bootstrap-based validation thus offered a robust framework that could effectively operate in the presence of real-world variability.

However, the use of experimental data, while essential to validate the method under practical conditions, also introduces a limitation: the intrinsic complexity of field measurements prevents a systematic and controlled assessment of the sensitivity of PCA with respect to different levels of structural damage. In other words, when dealing with real infrastructures, the damage scenarios cannot be finely tuned or precisely isolated, making it difficult to quantify how the method responds to incremental or localized deteriorations.

To overcome this limitation, the present work introduces a numerical study specifically conceived to reproduce controlled scenarios and to isolate the effects of damage from other sources of variability. The objective is to create an artificial yet representative environment where the detectability of damage can be investigated in a systematic and reproducible manner. In this framework, PCA is employed not as a direct damage index, but as a fundamental tool for dimensionality reduction, enabling the construction of a compact representation of the structural response space. Damage detection is then formulated as a problem of subspace comparison, in which the response of the intact structure defines the reference configuration, while the response of the damaged structure is projected onto a different subspace. The divergence between these subspaces, measured in terms of principal angles, provides the diagnostic evidence needed to identify and evaluate damage.

II. METHODOLOGY

A. *Numerical model*

The objective of this study, pursued through a numerical approach, is the development of a diagnostic tool capable of providing a quantitative and statistically supported assessment of structural damage, while also expressing its severity through a numerical parameter. The procedure relies on the analysis of the structural dynamic response, expressed in terms of acceleration time histories, which represent the most direct and informative signal for vibration-based diagnostics.

To establish a rigorous benchmark, the method was first calibrated on a controlled reference system, whose mechanical properties are well defined and unaffected by environmental disturbances. It is worth noting that the numerical signals used in this study do not include any additive measurement noise.

While this choice isolates the effect of stiffness degradation and allows the subspace-based indicator to be evaluated under idealized conditions, it also represents a limitation of the present simulation framework.

For this purpose, a finite element (FE) model of a simple yet spatially representative structural configuration was implemented. The model was subjected to excitation by means of an accelerogram characterized by variable amplitude and frequency content, and the resulting acceleration responses at selected nodes were extracted and organized for subsequent statistical processing.

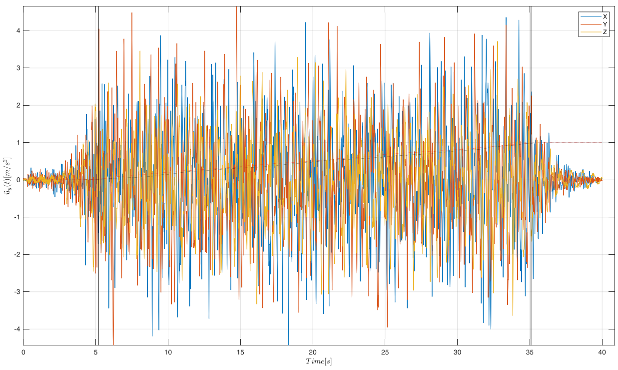

Fig. 1 - Input accelerogram - earthquake record

The numerical model consists of a slender vertical structure resting on a rigid base. This configuration was selected as a representative and simplified case study, suitable for reproducing the fundamental vibration modes in the two principal directions, x and y. The structural member was assigned inertial properties equivalent to those of a standard steel profile commonly available on the market. The discretization was performed using four beam-type finite elements and five nodes, each characterized by six degrees of freedom. This simplified configuration serves as a preliminary reference model, conceived to test and validate the proposed diagnostic procedure under controlled conditions.

The base node of the vertical structure was connected to a three-dimensional truss support system. This secondary structure did not influence the analysis results, as the focus was placed on the acceleration response matrices corresponding to the five nodes of the slender column. Although the support could have been simplified as a single rigid vertical member fixed at the base, a trussed configuration—commonly employed in the electromechanical field—was preferred to provide the overall model with a realistic geometrical representation.

The numerical investigation was conducted through four consecutive stages, each designed to simulate a progressive damage scenario under controlled conditions. In the first stage, the elastic modulus of steel (E=210,000 MPa) was adopted, representing the undamaged reference state of the structure. This configuration established the baseline dynamic behaviour and served to calibrate the PCA-based diagnostic framework on a healthy system, free of stiffness degradation.

In the subsequent stages, the elastic modulus of the second element of the slender portion—counting from the base—was systematically reduced in order to reproduce a localized loss of stiffness, corresponding to a gradual evolution of damage. The choice of the second element as the damaged component was intentional: its position ensures that the effects of stiffness reduction are transmitted through the structure without directly involving the boundary conditions, thus allowing a more realistic propagation of local damage to the global dynamic response.

The adopted sequence of stiffness reductions was defined as follows:

- **Stage 2:** E=100,000 MPa
- **Stage 3:** E=50,000 MPa
- **Stage 4:** E=25,000 Mpa

These values were selected to represent increasing levels of mechanical degradation—from moderate to severe—while still preserving the overall stability of the system. This progressive reduction provides a systematic framework for assessing the sensitivity of the diagnostic method to different degrees of damage severity. The induced stiffness losses modify the modal properties of the structure, producing measurable variations in the acceleration response, which are subsequently captured and quantified through the PCA and subspace comparison procedure.

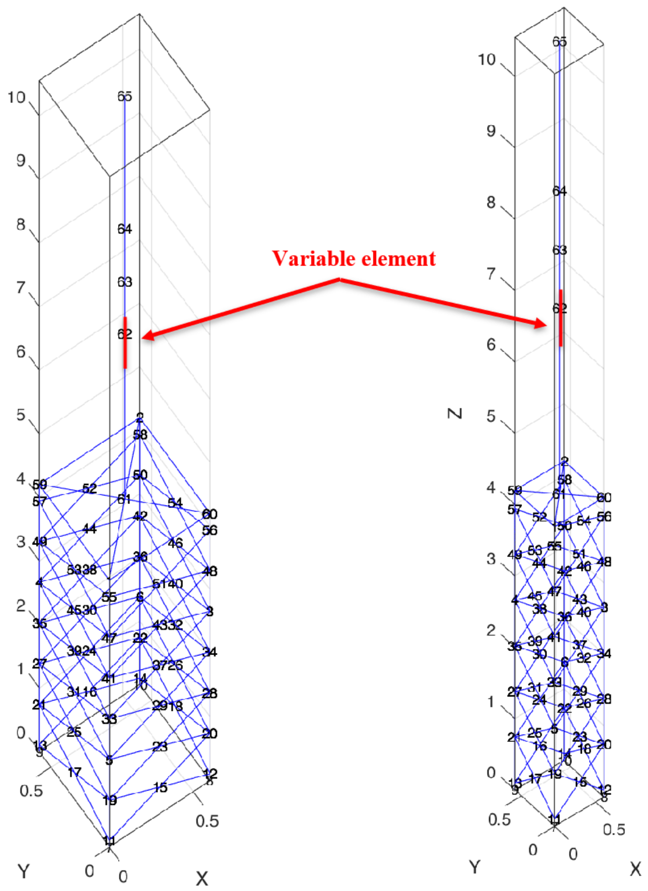

Fig. 2 - Finite Element Model (general view and mesh)

From the FE solver, acceleration responses in the x and y directions were extracted for all five nodes. The acceleration time histories consist of $n = 40\,000$ samples acquired at a sampling frequency of $f_s = 100$Hz, resulting in a total record length of 400 s. The corresponding FFT frequency resolution is

$$\Delta f = \frac{f_s}{n} = 0.0025 \text{ Hz}.$$

These data were subsequently organized into two matrices, respectively X and Y, each containing five columns corresponding to the acceleration time histories of the five nodes.

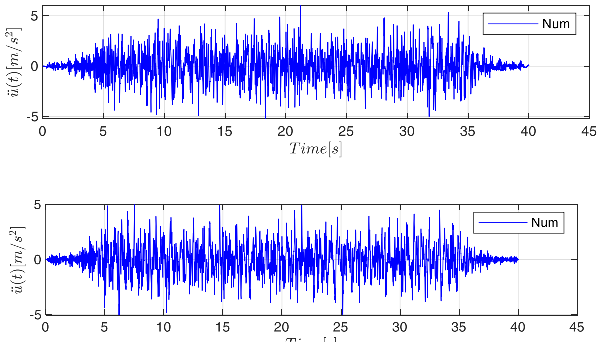

Fig. 3 – Typical accelerometric response of the points of the slender structure

Finally, the system was described by the matrix:

$$Z = X^2 + Y^2 \tag{1}$$

The adoption of configuration Z was motivated by the need to analyze a structural system whose dynamic behaviour is not governed by a single direction. This choice ensures a spatial response, closer to real structural conditions, and allows the proposed method to be tested under multidirectional dynamics, where modal interactions make damage detection more challenging and therefore more significant.

Indeed, many real structures, even when geometrically simple, do not exhibit a purely uniaxial dynamic response. The interaction between mass distribution, stiffness, and boundary conditions often produces coupled vibration modes that dominate the global behavior. For this reason, the numerical model was conceived with a simple geometry but designed to reproduce a spatially coupled dynamic response. Investigating an overly idealized structure vibrating in a single direction would have provided only limited insight, reducing both the generality and the practical relevance of the proposed diagnostic method.

For the intact configuration, the response data were arranged into the matrix $ZI=XI^2+YI^2$ with dimensions $(n\times p)=40{,}000\times 5$. The application of PCA to this matrix revealed that the first principal component accounts for approximately 90% of the system's total energy. On this basis, the optimal dimensionality reduction is achieved with k=1, leading to a one-dimensional subspace sZI that effectively represents the intact state of the structure.

B. Bootstrap distribution of the angle

To evaluate whether the observed differences between subspaces are statistically significant, a hypothesis test is formulated. The null hypothesis H_0 assumes that the subspaces represent the same dynamic behavior, implying that the maximum principal angle between them is zero. Since the sample distribution of principal angles under H_0 is not analytically known, we estimate it empirically using the Bootstrap method (Efron & Tibshirani, 1994), specifically employing a Moving Block Bootstrap (MBB) approach. This method accounts for temporal dependence within the vibration time series by resampling contiguous blocks rather than individual observations (Lahiri, 2003).

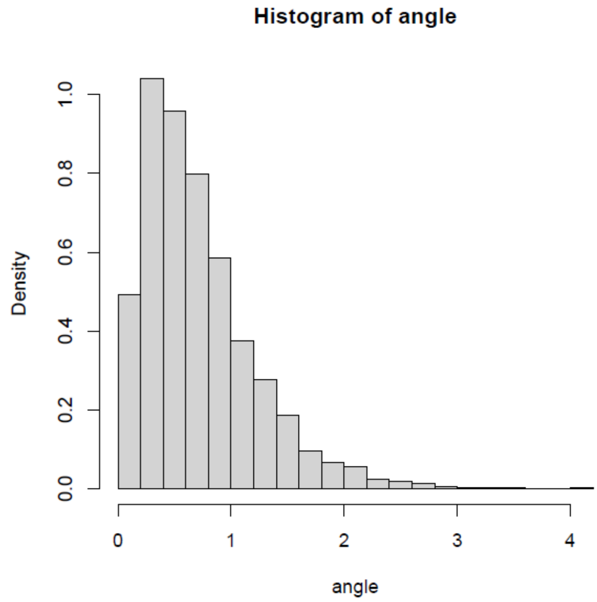

Fig. 4 - Sampling distribution of the angle θ

The figure shows the histogram of the sampling distribution of the angle θ, the corresponding to the maximum principal cosine, between sZI and the analogous samples generated, through moving-block bootstrap applied to time-series matrices of ZI.

III. RESULTS

The model is first modified by introducing a first level of damage (e.g., a localized stiffness reduction). The new matrix is:

$$Z1 = X1^2 + Y1^2 \tag{2}$$

with the same dimensions (nxp) = 40000 × 5. PCA shows that the first principal component explains about 96% of the energy. The comparison between sZI and sZ1 yields an angle of about 6°, significantly different from zero with confidence level 0.001.

With a second level of damage, more severe than the first, the resulting matrix is:

$$Z2 = X2^2 + Y2^2 \tag{3}$$

The comparison between sZI and sZ2 gives an angle of about 10°, also significantly different from zero.

Finally, introducing a third and more severe level of damage:

$$Z3 = X3^2 + Y3^2 \tag{4}$$

the PCA reduction (k = 1) and comparison with sZI return an angle of about 16°, again statistically significant.

Structural state	***Variance explained by 1st component***	***Principal angle vs. intact system***	***Statistical significance***
Intact (ZI)	~90%	-	-
Damage 1 (Z1)	~96%	~6°	$p < 0.001$
Damage 2 (Z2)	~95%	~10°	$p < 0.001$
Damage 3 (Z3)	~94%	~16°	$p < 0.001$

IV. DISCUSSION

The progressive increase in principal angles with damage severity (from 6° to 10° and then to 16°) highlights the capability of the proposed method not only to detect structural damage but also to quantify its extent. Within this framework, PCA serves as a tool for dimensionality reduction and information condensation, whereas the actual damage diagnosis is carried out by comparing the subspaces associated with the intact and the altered structural states. This strategy thus combines the efficiency of a compact representation with the discriminative power of subspace angles.

In practical applications, measurement noise would introduce additional variability in the acceleration time histories and could reduce the separation between the PCA subspaces associated with the intact and damaged states. In particular, noise tends to increase the variance explained by higher-order components and may partially obscure the changes induced by low-severity damage. However, the bootstrap-based statistical assessment adopted here can mitigate this effect by quantifying the confidence level associated with the observed subspace divergence. A future extension of this work will include controlled noise injection into the numerical model to evaluate the robustness of the proposed indicator under realistic ambient-vibration conditions.

V. ANALYSIS IN THE FREQUENCY DOMAIN

To further validate the sensitivity of the PCA-based diagnostic method, an additional investigation was carried out in the frequency domain, with the aim of correlating the statistical indicator obtained from subspace comparison with the physical variation of the dynamic properties of the structure. Indeed, any reduction in the stiffness of a structural element — simulated in this study by the progressive decrease of the Young's modulus in a single finite element — is expected to produce a corresponding decrease in the natural frequencies. By evaluating these changes alongside the PCA results, it is possible to establish a coherent relationship between the two diagnostic indicators.

The acceleration time histories of the four structural configurations (intact, first, second, and third damage) were processed through the Fast Fourier Transform (FFT) to identify the predominant frequency peaks associated with the fundamental vibration modes. As expected, the first two natural frequencies exhibited a clear and monotonic decrease as the damage severity increased. This trend directly reflects the loss of global stiffness resulting from the localized degradation introduced in the numerical model. The computed frequencies for the four damage scenarios are summarized in table below, together with their percentage reduction with respect to the intact configuration.

Structural state	***f_1 (Hz)***	***Δf_1 [%]***	***f_2 (Hz)***	***Δf_2 [%]***
Intact (ZI)	0.262	-	0.895	-
Damage 1 (Z1)	0.232	−11.4 %	0.850	−5.0 %
Damage 2 (Z2)	0.192	−26.7 %	0.800	−10.6 %
Damage 3 (Z3)	0.165	−37.1 %	0.737	−17.6 %

The progressive reduction of the natural frequencies provides an objective confirmation of the degradation simulated in the finite element model. When these variations are compared with the principal angles obtained from PCA subspace analysis (6°, 10°, and 16° for the three damaged states, respectively), a consistent and proportional trend emerges: the larger the reduction in stiffness (and thus in frequency), the greater the divergence between the subspaces representing the intact and damaged dynamic responses.

This dual evidence—statistical and physical—reinforces the robustness of the proposed diagnostic approach. PCA captures the structural changes through a geometric indicator in the feature space (the principal angle), while the FFT analysis links those changes to a measurable physical parameter, namely the shift in modal frequencies. Figure 5 illustrates this comparison by jointly plotting the evolution of the principal angle θ and the percentage reduction of the first two natural frequencies as functions of the damage level. To emphasize their common increasing trend, the frequency variation is represented on an inverted secondary axis, ensuring that all three curves consistently exhibit an upward trajectory with increasing damage severity.

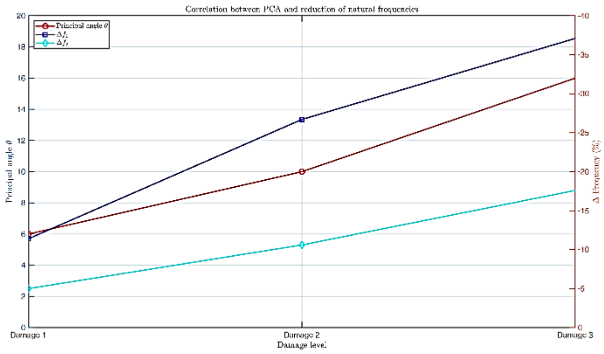

Fig. 5 - Comparison between the principal angle θ from PCA and the percentage reduction of the first two natural frequencies.

VI. CONCLUSIONS

This study has assessed, through a controlled numerical model, the capability of a PCA-based subspace comparison approach to detect and grade structural damage. The adoption of a moving-block bootstrap scheme provides a statistically robust basis for evaluating the significance of the observed subspace deviations. The numerical investigation complements earlier applications to real structures, offering a clearer perspective on the diagnostic potential of PCA and outlining several directions for future developments. Future work will incorporate simulated random noise to evaluate the reliability of the proposed indicator under operational conditions and to quantify its sensitivity in the presence of realistic disturbances. Additional extensions will include extending the analysis beyond the first principal component and integrating the approach with more advanced subspace-based techniques to further examine the robustness and applicability of the method.

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